

Questioning the exclusive focus on the Hu and Bentler norms in factor analysis:
Practice-oriented Likert scale indicators based on an analysis of 161 datasets

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Abstract

This article challenges the wide-spread practice of using Hu and Bentler's (1999) cut-off values ($CFI > .95$, $TLI > .95$, $RMSEA < .06$, $SRMR < .08$) as the sole indicator of good test fit. Two underlying assumptions are questioned: (1) that these criteria apply universally and (2) that very high CFI/TLI and low RMSEA/SRMR represent optimal fit. An exclusive focus on Hu and Bentler's criteria favors top-down test construction with narrow, pre-defined scales, potentially neglecting bottom-up approaches that identify items reflecting real-world complexities. This article proposes an alternative approach: compiling practice-based distributions of fit indices from existing tests. An analysis of 161 datasets revealed distinct high-probability regions for TLI, RMSEA, and SRMR, largely outside the Hu and Bentler thresholds and dependent on the statistical technique used for analysis. Based on the distributions, indicators are developed to differentiate between unifactorial, multifactorial, and hierarchical scales. The indicators are intended to complement, not replace, statistical model comparison by providing a practice-oriented perspective.

Factor analysis has become a core method in questionnaire construction and analysis (Rogers, 2024). Researchers want to know whether their scale is unifactorial, hierarchical, or multifactorial.

Types of scales

A unifactorial scale is a scale that measures one factor (trait), such as depression, optimism, face recognition, or perceived stress. The sum of the items is usually taken as the best estimate of a person's position on the trait, although there is debate as to whether factor loadings would be better (McNeish & Wolf, 2020; Widaman & Revelle, 2023). The advantage of sum scores is that they are comparable across studies; the limitation is that the scale ideally does not contain weak items.

A hierarchical scale is a scale that also measures a single trait, but the trait is assumed to contain several correlated components. The best known example of a hierarchical skill is intelligence: Even though all intelligence tests correlate positively with each other, some tests correlate more with each other than with other tests, giving rise to intelligence components, such as crystallized intelligence, fluid intelligence, verbal comprehension, perceptual reasoning, working memory, processing speed. The total score of a hierarchical scale gives information about a person's position on the trait being measured, but additional information can be obtained by looking at the scores on the constituting components. It may be argued that all unifactorial scales are hierarchical scales in disguise, because not all items are likely to correlate to the same extent with each other and the addition of more items would expose the hierarchical nature of the scale. The practical distinction between a unifactorial and a hierarchical scale is whether useful information is obtained by considering the values of the correlated skill components. For instance, the distinction between crystallized and fluid intelligence is helpful because older adults score lower on fluid intelligence tests than younger adults (McArdle et al., 2002), whereas the same is not true for crystallized intelligence up to a high age (Rönnlund et al., 2005; Salthouse, 2019).

A multifactorial scale measures several, more or less independent traits. There is no point in calculating a single scale score, as this score does not represent useful information. Only scores for the constituting traits are of interest. The best known example of a multifactorial scale is a personality test, where the test is assumed to measure five or six traits (extraversion, openness, ...) that are largely independent of each other (Marsh et al., 2013).

The typical approach to deciding which type a scale belongs to is to examine how well the data match the different models. Authors thus examine whether their data most closely match a single factor, a hierarchy of correlated factors, or a collection of largely independent factors (e.g., Coroiu et al., 2018; Extremera Pacheco et al., 2019; Gray et al., 2016).

The Hu and Bentler criteria for goodness of fit

Four measures of model fit are often mentioned: CFI, TLI, RMSEA, and SRMR. CFI (comparative fit index) and TLI (Tucker-Lewis index) are indices of goodness of fit and inform us how well the hypothesized model compares to the absence of structure in the data. The higher CFI and TLI the better. RMSEA (root mean square error of approximation) and SRMR (standardized root mean squared residual) are measures of deviation and indicate how far the hypothesized model is from a perfect model. The lower RMSEA and SRMR, the better.

Values for good scales commonly given are CFI > .95, TLI > .95, RMSEA < .06 and SRMR < .08. These values stem from a study by Hu and Bentler (1999) and have more or less become the standard against which new scales are evaluated by reviewers, editors and potential users. What people often do not realize is that these criteria are based on a small simulation study with a single estimator (maximum likelihood) and a design consisting of three correlated factors (correlations of .3, .4 and .5) each measured with 5 items that loaded .70-.80 on their factor and for the simple model did not load on the other two factors, and in the alternative complex model had one cross load on another factor. When different designs are simulated, the criteria change (McNeish, 2023). Another limitation of the criteria of Hu and Bentler (1999; or those of McNeish, 2023) is that they were obtained in simulation studies, where all non-essential characteristics matched the statistical assumptions underlying the analysis. Such an ideal situation is unlikely in practice.

Negative aspects of an exclusive focus on the Hu and Bentler criteria

Strict criteria need not be bad if they do not have negative effects. However, such effects cannot be ruled out because the Hu and Bentler criteria for test construction are severe: high factor loadings, no cross-loadings, and no correlated residuals between items (i.e., no correlations between items that cannot be explained by the item loadings on the factors; Cole et al., 2007). These are strong constraints on scale construction that may skew the outcome. For example, the number of items per factor is likely to be small (because they are hard to find) and authors will tend to omit methodological controls in the item construction. Regarding the latter, including inversely scored items in a scale (e.g., "I am bad at resisting temptation" instead of "I am good at resisting temptation") is known to have negative effects on model fit (Garcia-Pardina et al., in press; Paap et al., 2024). Reasons for this are that negatively worded items rarely have exactly the opposite meaning from positively worded items (Paap et al., 2024) and that people differ in their tendency to agree with statements. Some people are more likely to agree with *both* the statement "I am bad at resisting temptation" and the statement "I am good at resisting temptation" than other people. The latter is known as the acquiescence bias (Kam & Meyer, 2015). To reduce any noise that reduces model fit, it is tempting for authors to omit items with reverse coding. However, if you do not control for acquiescence bias, you run the risk of misinterpreting correlations between factors because they may be due to response bias.

Because it is difficult to create bottom-up scales that meet all of Hu and Bentler's (1999) criteria, the temptation is to work top-down. First, the researcher outlines the model behind a scale and then creates items to measure each component. For example, if a group of researchers believe that a particular trait is hierarchical and has three components, they may create items that meticulously measure each component. As long as the number of items is fairly small (say three to four) and not worded too similarly (so that there are no correlated residuals between item pairs), it is possible to create a scale that measures the hypothesized model and meets Hu and Bentler's (1999) quality criteria if the right analysis package is used. The key question with such a scale, however, is whether it measures something meaningful in people or simply reflects meaning similarities between the words and phrases used by the researchers. As Cooper (2024, p. 3) cautioned: "... there is no guarantee that the traits measure real characteristics of people. They may instead be social constructions ... They may describe characteristics which we *think* exist, but which have no basis in reality".

Another problem for narrow scales is that they can be vulnerable to jingle or jangle fallacy. A jingle fallacy is when two tests claim to measure the same characteristic but have little content overlap. Fried (2017) noted that there is surprisingly little content overlap between seven commonly used scales to measure depression. Similar observations have been made for other screening questionnaires. The jangle fallacy occurs when researchers introduce a new concept that is in fact almost perfectly similar to an existing trait (Hodson, 2021). For example, Ponnock et al. (2020) argued that grit is not sufficiently different from conscientiousness to warrant separate status.

To avoid jingle-jangle fallacies, Cooper (2024) advocated a bottom-up approach in which a problem space is charted as broadly as possible and existing correlations are discerned and explained. Such an approach was followed by Altgassen et al. (2024), who noted that mindfulness questionnaires differed greatly from one another (jingle) while providing little meaningful incremental criterion validity relative to other factors such as personality. However, broad, bottom-up questionnaires are unlikely to meet the Hu and Bentler criteria, because the wider a scale becomes the lower the factor loadings.

Finally, an exclusive focus on statistical fit encourages researchers to use analysis techniques that bring the fit measures as close to the criteria as possible.¹ One thing that helps is the use of a bifactor model. This is a model that consists of one general factor, which affects all items, and a number of independent factors that affect subsets of items. Bifactor models have been shown to fit datasets better than other models, even when those datasets were generated by an algorithm that did not follow a bifactor model (Greene et al., 2019). Another temptation is to try different analysis techniques to find the one that best approximates the statistical criteria.

How well do existing tests approach the Hu and Bentler criteria?

A look at the literature suggests that Hu and Bentler's criteria are often not met with existing, well-established tests based on Likert scales. So, an exclusive focus on those criteria may do more harm than good. To investigate the impression more systematically, I decided to look for some 150 data sets that could be analyzed in a uniform manner. Roughly one-third of these measures (N = 52) were unifactorial, one-third were hierarchical (N = 59) and one-third were multifactorial (N = 50). Datasets were mostly found on the open science framework (<https://osf.io/>), although a recurring problem was that authors made available only sum scores per test and not the raw scores needed for factor analysis.

I also made an effort to contact authors if the raw data were not available. In particular I contacted the authors indicating that “The raw data supporting the conclusions of this article will be made available by the authors without undue reservation” or “The dataset that supports the findings of this study will be available from the corresponding author upon reasonable request”. In total, I sent 74 requests, and obtained 24 positive responses (36%), a percentage in line with previous studies (Hamilton et al., 2023; Tedersoo et al., 2021; Vanpaemel et al., 2015). Nine more authors responded but in the end did not send data (arguably because the request was not thought to be reasonable enough).²

¹ Or they are forced to do so by the reviewers.

² Authors not making data available may ponder the fact that they just missed a citation to their article. Editors may also want to take this into consideration.

Table 1 gives an overview of the datasets used. They are available on https://osf.io/e3dmh/?view_only=2196558e8c3e4ae3bfff6b60bf0b9464, together with the analysis code. I tried to limit the number of datasets to one per study, except for four instances where the information was so difficult to find that I included two datasets based on the same sample of participants. The minimum number of participants of a study was 85; the maximum was 19,521. The number of response alternatives on the Likert scale varied from 3 to 11. The number of items per scale varied from 7 to 100.

Table 1: Datasets analyzed in the present study

Type of scale	N datasets	References
Unifactorial		
Compliance COVID	4	Blackburn et al. (2022; Spain, Finland, Brazil, Slovakia)
Compulsion	2	Muela et al. (2023; two samples)
Depression (BDI-II)	4	García-Batista et al. (2018), Rabeya et al. (2023), Sandoval-Lentisco et al. (2024), Smith et al. (2016)
Face recognition (CFMQ, CFMT, GFMT, PI20)	6	Dal Lago et al. (2023), Estudillo (2021), Estudillo & Wong (2021), Grabman & Dodson (2024), Kramer (2023), Nørkær et al. (2023)
Handedness	1	Dragan et al. (2022)
Interoception (IAS)	7	Brand et al. (2023; 3 samples), Campos et al. (2021), Gaggero et al. (2021), Murphy et al. (2020), Todd et al. (2022)
Loneliness (UCLA)	4	Lardone et al. (2020), McDanal et al. (2021), Panayiotou et al. (2023), Weinstein & Nguyen (2020)
Optimism	2	Coelho et al. (2018)
Perceived stress COVID	4	Blackburn et al. (2022; Arabic, Bulgaria, Germany, Japan)
Self-control (BSC)	6	Paap et al. (2024; 4 samples), Sjästad & Ekström (2021; 2 samples)
Social Dominance (SDO)	7	Fleeson et al. (2023, 2 samples), Perkins et al. (2022), Roets (2019; 2 samples), Simon et al. (2023; 2 samples)
Trust COVID	5	Blackburn et al. (2022; Denmark, Honduras, Sweden, Turkey, UK)
Hierarchical		
DASS-21	2	Chen et al. (2023), Thiagarajan (2022)

Emotional intelligence (AES, IRI, TEIQue, W&L)	13	Anglim et al. (2020), Bainbridge et al. (2022), Brienza et al. (2018), Briganti et al., (2018), Demetrovics et al. (2022; also Kun et al., 2022), Martingano & Konrath (2022), Perazzo et al. (2021), Pérez Díaz & Petrides (2021), Radeva (2023), Robinson et al. (2023), Rúa (2019), Stosic et al. (2022)
Emotion regulation COVID	3	Blackburn et al. (2022; USA, Guatemala, Russia)
Hope	1	Boutilier (2020)
Impulsivity (BIS, UPPS)	12	Billieux et al. (2021), Demetrovics (2013; also Kiraly et al., 2019), Eben et al. (2020), Flayelle et al. (2022), Haines et al. (2020), Littrell et al. (2020), Meule (2020), Racine et al. (2024), Steiner & Frey (2021), Wuellhorst et al. (2023), Zsila et al. (2020)
Interoception (MAIA)	5	Desmedt et al. (2022), Ferentzi et al. (2021), Randelović et al. (2023), Reis (2017), Rogowska et al. (2023)
Moral values COVID	4	Blackburn et al. (2022; Czech Republic, Colombia, Estonia, Portugal)
OCD	3	González et al. (2011), Ignatova & Onchev (2023), Ozcanli et al. (2020)
Personality disorder (Dark triad)	1	Nielsen et al. (2023)
PTSD (PCL-5)	3	Ashbaugh et al. (2016; 2 samples), Orovou et al. (2022)
Quality of life	2	McConachie et al. (2018), Rogers (2022)
Sense of coherence	2	Lelek-Kratiuk & Szczygieł (2022), Zajenkowski & Szymaniak (2021)
Sexual desire (SDI-2)	1	Jones & DeBruine (2018)
STAI-T	3	Joshi et al. (2023), Sundelin & Holding (2022), Werner et al. (2022)
Well-being	1	Ishii et al. (2022)
Multidimensional		
Beliefs and values	1	Barnby et al. (2019)
BIS/BAS scale	6	Binter et al. (2023), Dierickx et al. (2022), Jonker et al. (2022), Khaliq et al. (2023), Molenda et al. (2022), Rutten et al. (2016)
Mindfulness (FFMQ)	15	Karl et al. (2020; 15 different countries)
Moral Foundations	2	Harper & Rhodes (2021), Zakharin & Bates (2021)
PANAS	1	Díaz-García et al. (2020)
Perfectionism	4	Kaçar-Başaran et al. (2022), Linnett & Kibowski (2020), Mackinnon et al. (2019), Workye et al. (2023)

Personality (16PF, BFAS, BFI2, BFPTSQ, HEXACO, IPIP, RIASEC, TIPI)	13	Andrejević et al. (2022), Coelho et al. (2018; 2 samples), Gallardo-Pujol et al. (2022), Gallagher et al. (2023), Henry et al. (2022), Mezquita et al. (2019; 2 samples), Openpsychometrics (2 samples), Silvia et al. (2009), Vermeiren et al. (2023; 2 samples), Zajenkowski & Szymaniak, (2021)
Personality disorders (FFMI, ICD-11, LPFS-BF, LSRPS, SRP4)	5	Carnovale et al. (2020), Grabovac & Dinić (2022), Knack et al. (2021; 2 datasets), Natoli et al. (2022)
Reinforcement sensitivity (RST-PQ)	3	Nuel et al. (2022), Satchell et al. (2019), Sozer et al. (2022)
Time perspective	3	Akirmak (2021), Bodecka et al. (2021), Ceccato et al. (2021)

R packages used and analyses done

Two R packages are often used for factor analysis: Psych (Revelle, 2017) and Lavaan (Rosseel, 2012). The Psych package is particularly strong for exploratory factor analysis (EFA). This type of analysis is used when researchers want to know how many factors are needed to account for the correlations between the items. Lavaan is particularly strong for confirmatory factor analysis (CFA), when researchers want to know how well their data match the theoretical model behind the test.

The basic command in Psych is `fa(dataset, Nfactors)`. This command informs the user what the item loadings are for a unifactorial model [`fa(dataset, 1)`], a model with two factors [`fa(dataset, 2)`], or any other number of factors. The default analysis (at least for package version 2.3.6 I used) makes the following choices: 1) correlations between items are Pearson correlations, 2) ordinary least squares (OLS) is used to find the minimum residual (minres) solution, and 3) oblimin rotation is used, so that correlations between factors are allowed. These are the choices that the author of the package thinks are best for an initial factor analysis.

Pearson correlations are not optimal for Likert scales because such a scale is ordinal and the distributions of responses to items may be skewed (e.g., an item where most people agree or disagree). Pearson (1909) already described how answer probabilities can be used to estimate the underlying normal distribution, which then makes it possible to calculate the correlation between the underlying distributions. Such correlations have since become known as polychoric correlations. Typically, polychoric correlations for Likert scales are higher than Pearson correlations, allowing for more variance to be explained in the data and giving greater weight to the first factor.

To run a factor analysis with polychoric correlations in Psych, the following commands must be used:

```
rho <- polychoric(dataset)$rho
fa(rho, Nfactors, n.obs=nrow(dataset))
```

Polychoric correlations usually work well, but may give anomalous values if there is not much variance in the responses to some items, or if the estimates are based on small samples. In that case, the package

Psych gives a warning and applies some form of smoothing in some cases. The author of the Psych package further refuses to give polychoric correlations if the number of response options given is not the same for all items or if the number of answer alternatives is greater than eight. For these two cases it is possible to calculate polychoric correlations with the package EGAnet (Golino & Christensen, 2023). Because polychoric correlations involve more warnings than Pearson correlations, I conducted both analyses.

In lavaan (version 0.6.17), a model must be defined. For a unifactorial scale, all items are assumed to load on a single factor. For a hierarchical or multifactorial scale, multiple factors are defined and items must be assigned to the specific factors. For example, for a personality test, five or six different traits (factors) are distinguished and items are assigned to the factor they are supposed to measure.

Two methods are recommended for Likert scales. The first is MLR (maximum likelihood robust). This method uses a workhorse from statistics (maximum likelihood estimation) and adjusts it for situations where the variables are not normally distributed. The second method is DWLS (diagonally weighted least squares). This method is based on polychoric correlations and weights correlations relatively to their informativeness for the underlying structure (Muthén & Muthén, 2007). It results in higher estimates of explained variance and model fit (CFI and TLI), at the expense of higher variability (RMSEA and SRMR), more warnings, and incidental results that differ significantly from the other models, making it feel like an analysis on steroids. In Lavaan, the procedure is used when the ordered=TRUE option is selected.

All in all, four analyses were performed: Psych.pearson, Psych.polychoric, Lavaan.MLR and Lavaan.ordered. The results were saved regardless of whether a warning was given, so that we would get an idea of how the different analyses do in a pipeline when all warts are included.

Unifactorial scales

As described in the previous section, the following commands were used (see the R code on osf for full details).

```
Psych: EFA_model1 <- fa(dataset,1)
Psych: EFA_model2 <- fa(polychoric(dataset)$rho,1,n.obs=nrow(dataset))
Lavaan: CFA_model1 <- cfa(UNImodel, data = dataset, estimator="MLR", std.lv=TRUE)
Lavaan: CFA_model2 <- cfa(UNImodel, data = dataset, ordered=TRUE, std.lv=TRUE)
```

Figure 1 shows the distribution of the values obtained for three Hu and Bentler variables: TLI, MSREA, and SRMR. To these I added percentage of variance explained, as this usually is an informative measure in model fitting. I did not calculate CFI, as this is not given by default in the psych package. For lavaan I used the scaled non-robust measures. For comparison, I also plotted the Hu and Bentler norms, which are seen as the current standards.

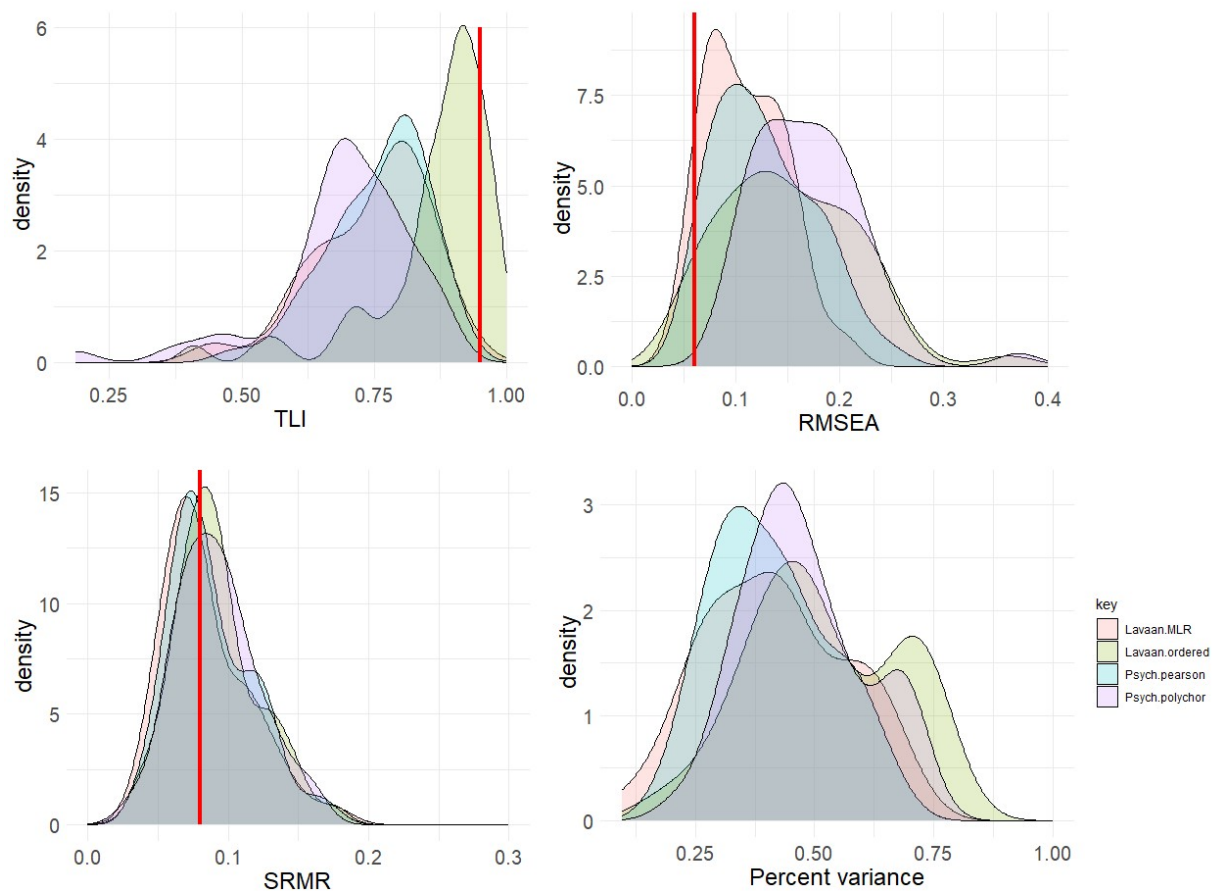


Figure 1: Values for TLI, RMSEA, SRMR and Percentage explained variance obtained in 52 analyses of datasets meant to represent a unifactorial scale.

Two findings are noteworthy in Figure 1. First, very few tests meet Hu and Bentler's quality criteria. The situation is best for SRMR, where the modes of the distributions match the putative criterion. The situation is dire for TLI and RMSEA, where almost none of the existing scales meet the putative criteria.

The second noteworthy finding is that results differ as a function of the analysis done. This is particularly true for TLI and RMSEA. Lavaan.ordered gives by far the highest TLI values, Psych.polychoric the lowest. Psych.polychoric gives the highest values of RMSEA, Lavaan.MLR the lowest. Psych.polychoric explains more variance than Psych.pearson.

Multifactorial scales

Figure 2 shows the outcome of a similar analysis for the 50 datasets of multifactorial scales. For these analyses, it is important to know that the Lavaan analysis is much stricter than the Psych analysis. In Lavaan, each item is assigned to the specific factor it is expected to measure. In Psych only the number of factors is restricted; items are free to load so that the dataset is fittest best. Lavaan tests for measurement invariance, whereas Psych only tests for (weak) factorial invariance.

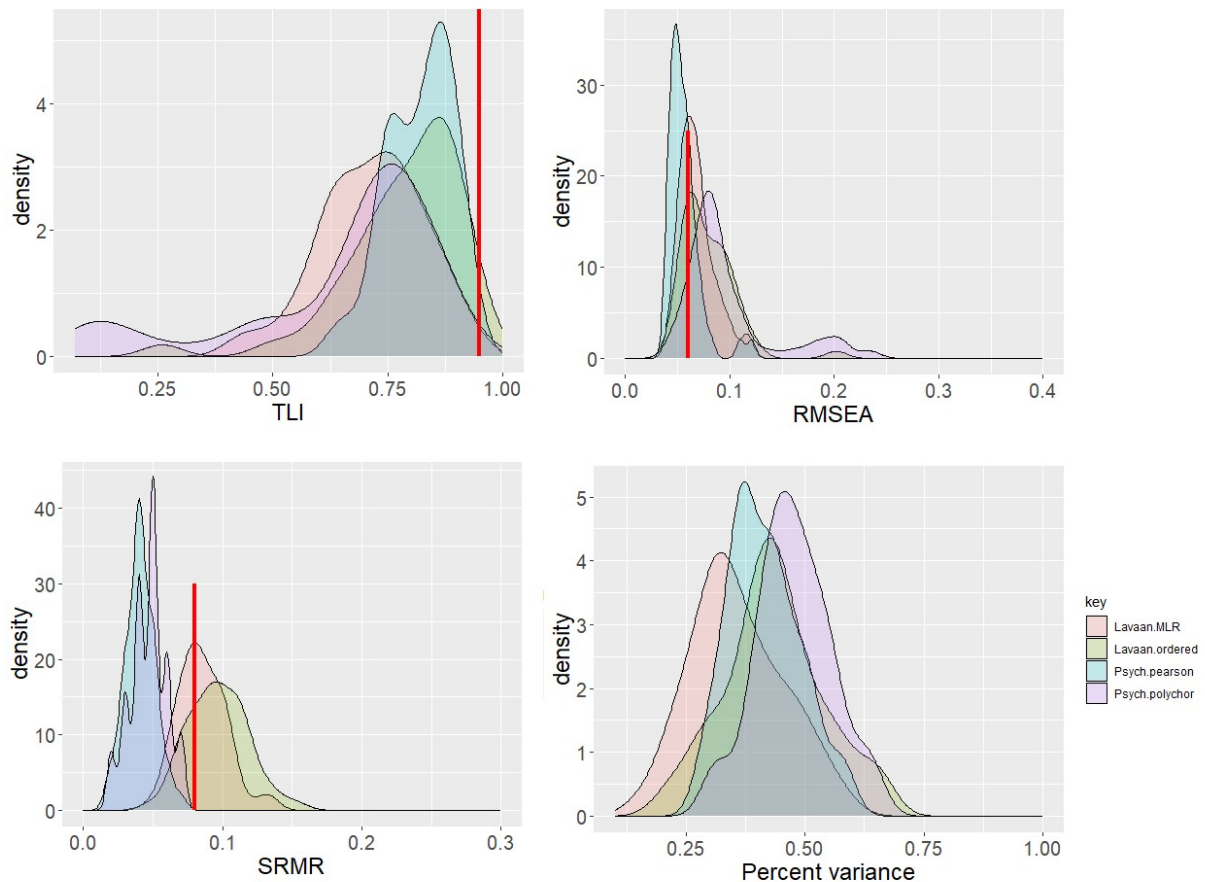


Figure 2: Values for TLI, RMSEA, SRMR and Percentage explained variance obtained in 50 analyses of datasets meant to represent a multifactorial scale.

For multifactorial scales, the situation is reasonably good for RMSEA and SRMR, certainly with the package Psych (which does not constrain the factor to which an item belongs). The situation is particularly good for Psych.pearson. Psych.polychoric occasionally goes awry (low TLI, high RMSEA), in line with the regular warning messages it elicited (in about one quarter of the analyses). TLI remains much lower than assumed on the basis of the Hu and Bentler norms. Psych.pearson does best, opening the likelihood that authors will report different fit measures for unifactorial and multifactorial scales.

SRMR makes the biggest distinction between Psych (no measurement invariance) and Lavaan (measurement invariance).

Hierarchical scales

Figure 3 gives the outcome of the 54 datasets that are thought to measure hierarchical traits.

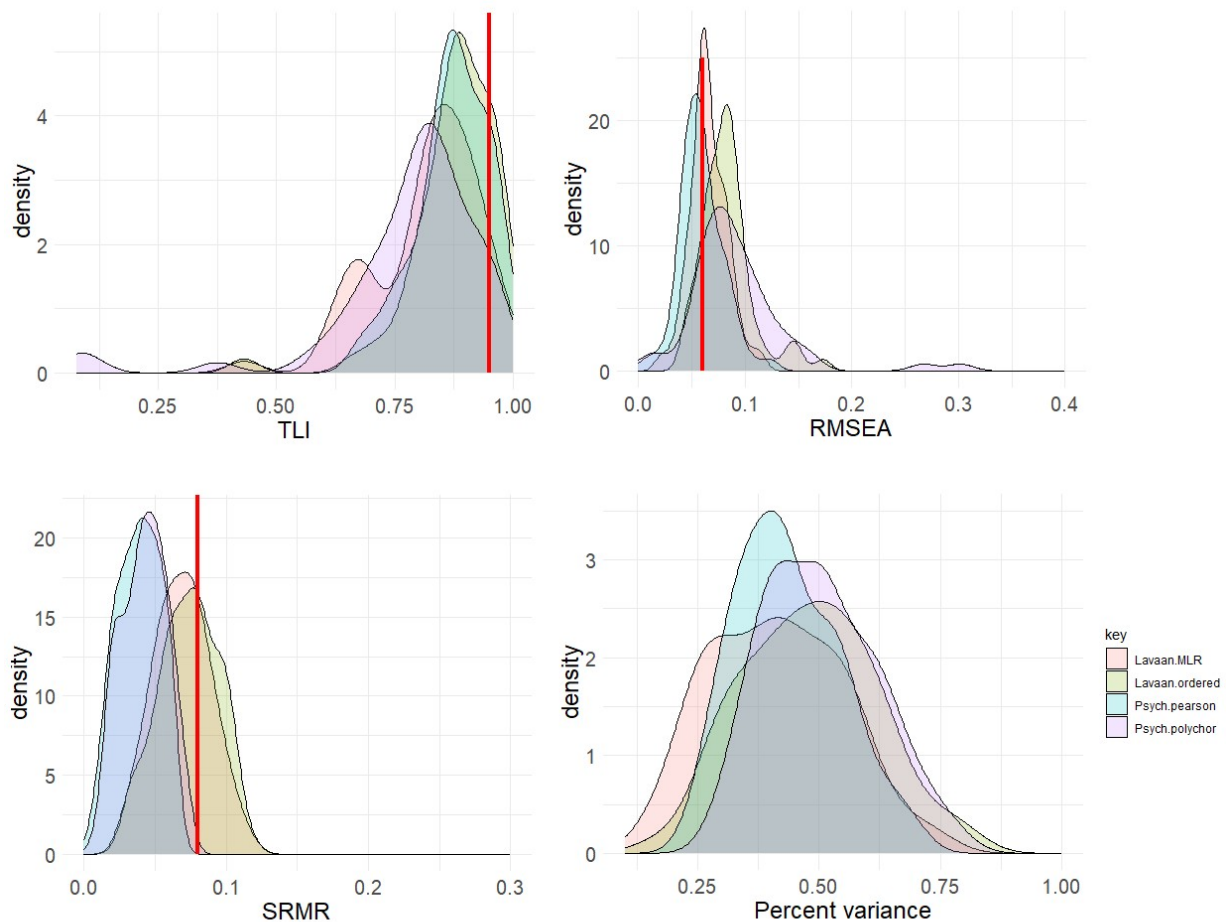


Figure 3: Values for TLI, RMSEA, SRMR and Percentage explained variance obtained in 54 analyses of datasets meant to represent a hierarchical scale.

As can be seen in Figure 3, hierarchical scales look the best in terms of the model fit, still being well short of the Hu and Bentler criteria. In particular Psych.pearson scores well, except for the criterion percentage of variance explained. SRMR again makes the biggest distinction between Psych and Lavaan.

Are the criteria informative at all?

If the Hu and Bentler criteria cannot be met, does this mean that measures of fitness are of no use? This need not be the case. To answer this question, we need to know how measures perform when a model is mis-specified. Hu and Bentler simulated this by having one cross-loading per item. In practice, however, researchers are primarily interested in whether the scale they have developed measures one or more factors, and whether these factors correlate sufficiently to form a hierarchy.

Fit to unifactorial model

For a unifactorial scale, researchers particularly want to make sure that the scale is not multidimensional. Thus, they would like to see how the different measures perform when the scales are forced unifactorial. Figures 4 and 5 show what this looks like for Psych.pearson and Lavaan.MLR. The figures for Psych.polychoric and Lavan.ordered look about the same, but these analyses raise more frequent warnings.

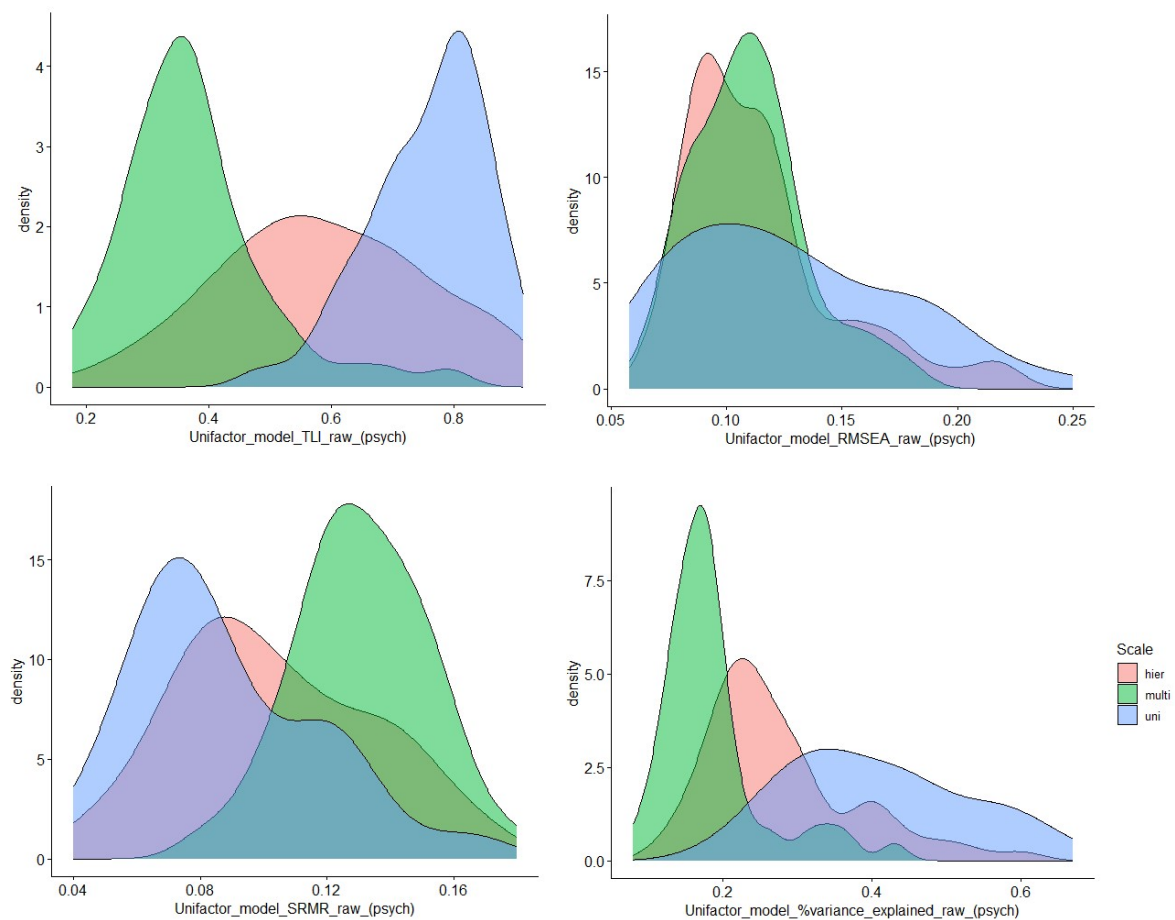


Figure 4: Values Psych.pearson for TLI, RMSEA, SRMR and Percentage explained variance as a function of scale type when the statistical analysis is based on the assumption that the scale is unifactorial.

TLI makes a clear distinction between unifactorial and multifactorial scales, with the caesura slightly below TLI = .6. The hierarchical scales have most values between .45 and .7. A similar distinction can be seen for SRMR and percent variance explained but, surprisingly, not for RMSEA. A very similar picture emerges from Lavaan.MLR (Figure 5), with borders around .45 and .6 for TLI.

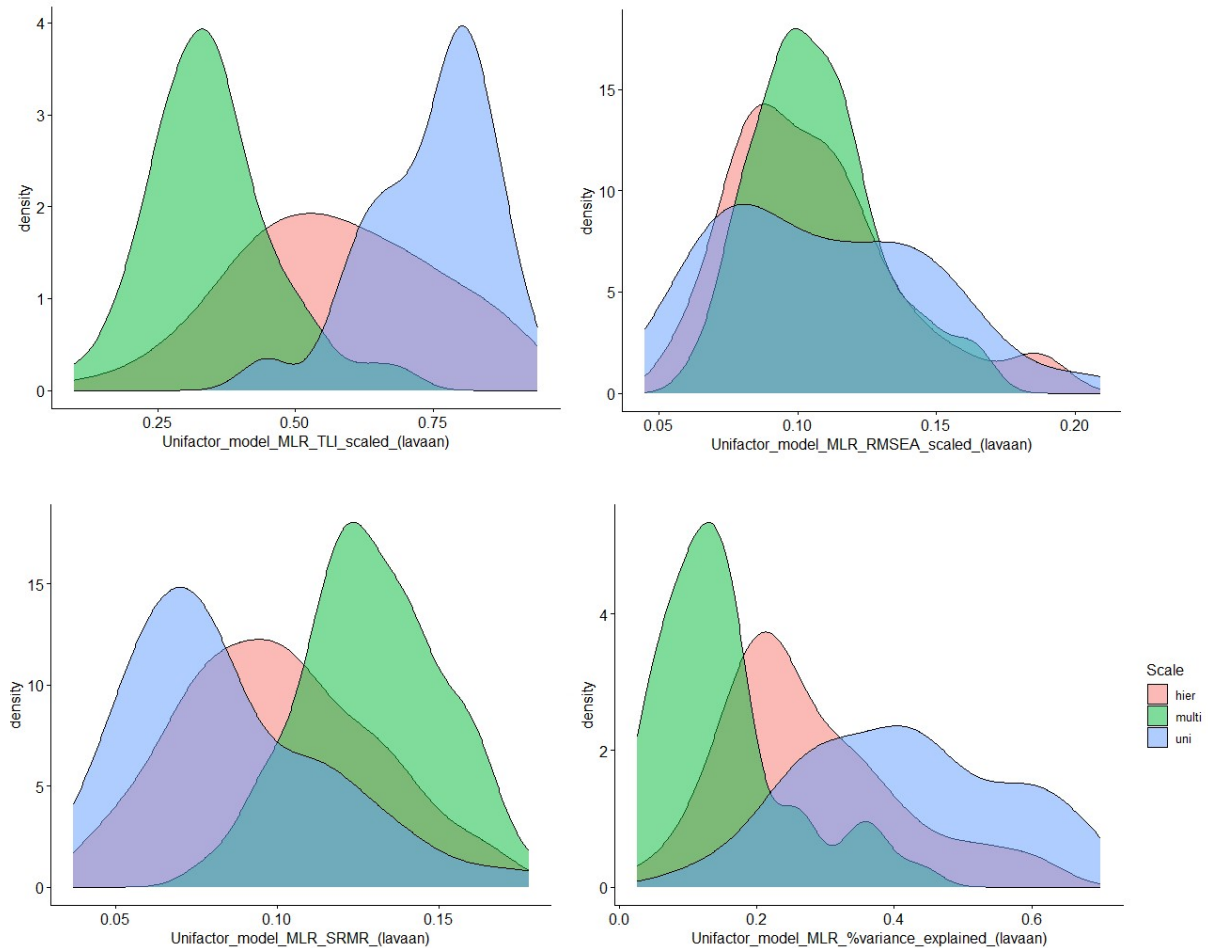


Figure 5: Values Lavaan.MLR for TLI, RMSEA, SRMR and Percentage explained variance as a function of scale type when the statistical analysis is based on the assumption that the scale is unifactorial.

Fit to multifactorial model with orthogonal factors

We can perform an analogous analysis assuming a multifactorial scale by forcing the statistical model to include orthogonal factors (i.e., factors that do not correlate with each other). Now the model fit measure must favor a multifactorial scale and penalize a unifactorial scale. A challenge here is how to break down a unifactorial scale into multiple factors. Three approaches were combined. First, if there was a good review article that argued that one of the unifactorial scales contained more than one factor,

this evidence was used to define the contributing factors. Second, if such information could not be found and if I had more than one study for a scale, I calculated the most likely structure from all the data and used this as the structure of the scale. Finally, if none of these conditions were present, I simply took the structure suggested by an EFA analysis. Figure 6 shows the result of Lavaan.ordered, when the following command was used:

```
CFA_model1 <- cfa(MULTImodel, data = dataset, estimator="MLR", std.lv=TRUE,
  orthogonal=TRUE)
```

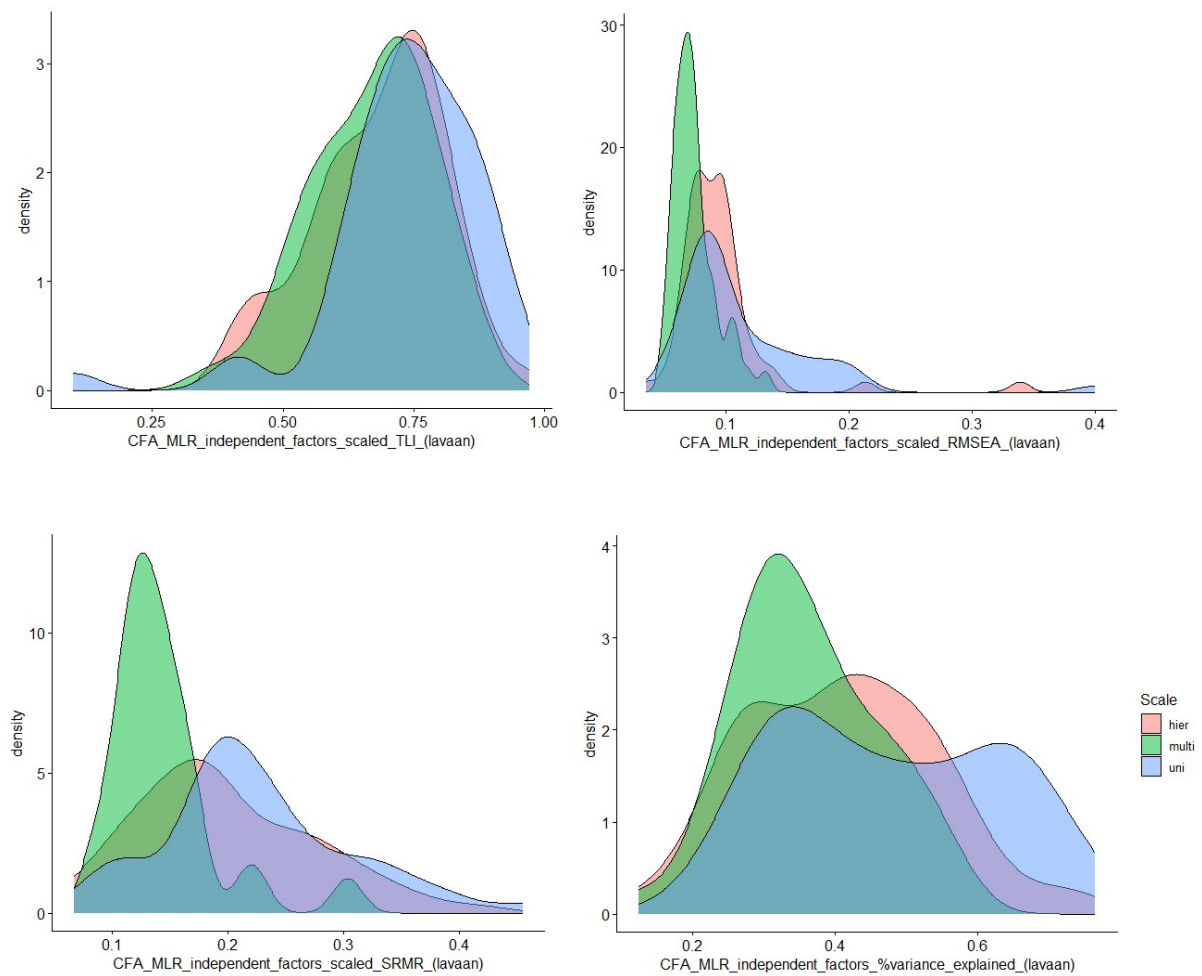


Figure 6: Values Lavaan.MLR for TLI, RMSEA, SRMR and Percentage explained variance as a function of scale type when the statistical analysis is based on the assumption that the scale is influenced by a number of independent factors.

Surprisingly, in this analysis TLI make virtually no difference between the scales. The clearest difference is found for SRMR: SRMR < .18 is good evidence for a multifactorial scale. SRMR > .2 is good evidence for a unifactorial or hierarchical scale.

Correlations between factors in a multifactorial model

The main difference between a hierarchical and a multifactorial design is that a hierarchical model assumes substantial correlations between factors, while a multifactorial model assumes low correlations between factors. A unifactorial scale should follow the pattern of a hierarchical model and have high correlations between the "factors." So it may be interesting to look at the average correlation between factors in the various analyses. Since correlations between factors can be either positive or negative (depending on how the factor is defined), we need to look at the absolute values of the correlations. The result is shown in Figure 7.

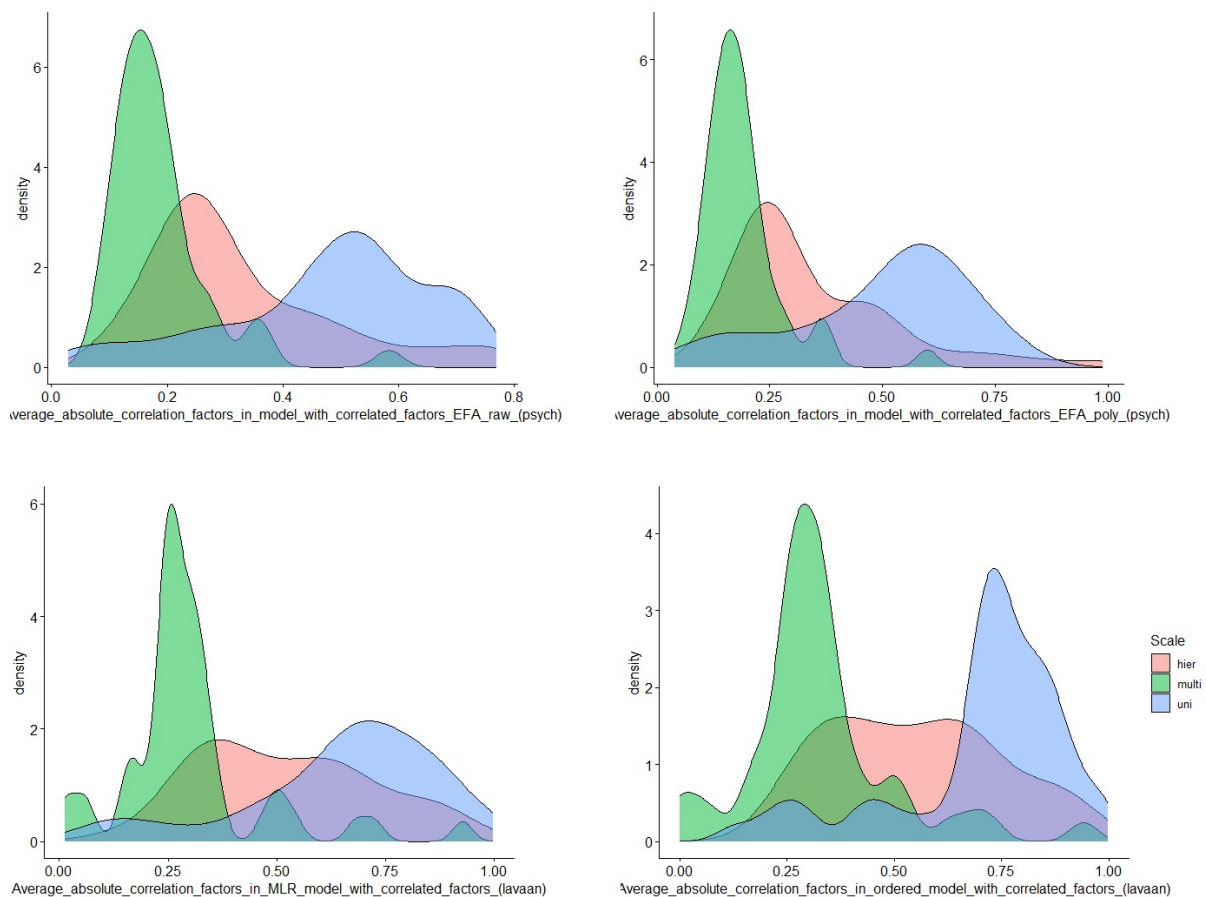


Figure 7: Absolute correlations between the factors of a multifactorial model as a function of the type of scale. Upper panel: Psych.pearson, Psych.polychoric. Lower panel: Lavaan.MLR, Lavaan.ordered.

The correlations values differ as a function of the measure, with higher values for polychoric correlations (right side) than for Pearson correlation (left side). Correlations between factors of hierarchical models are much lower in Psych than in Llavaan, which is to be expected given that in Psych items are distributed free over factors to make the factors as distinct as possible. A correlation lower than .4 in Llavaan.MLR is good evidence for a multifactorial scale or hierarchical scale; a correlation higher than .7 in Llavaan.ordered is good evidence for a unifactorial (or hierarchical scale).

Change from a unifactorial to a multifactorial model

Figures 4-7 are just four examples of variables that can be calculated. Another successful one I tried was the change in the percentage of variance accounted for when going from an unconstrained unifactorial model in Psych to an unconstrained multifactorial model. The reasoning is as follows. If a scale is truly unifactorial, being able to add additional factors to the model will not make much difference. However, being able to add additional (correlated) factors to the model will improve the fit a lot for a multifactorial model and also to some extent for a hierarchical model.

We can operationalize the change in fit by looking at the percentage change relative to the unifactorial model. Suppose a unifactorial model has a TLI of .6 for the scale we are examining. A 3-factor EFA has a TLI of .8. This makes a difference of $(.8 - .6) / .6 = .33$. If the TLI of the multifactorial model is lower than that of the unifactorial model, we get a negative value. Such a value is possible for a truly unifactorial scale that is better described by a single-factor model than by a multifactorial model, but in general we would expect the changes to be positive (the multifactorial model gives a higher value of the TLI than the unifactorial model).

Figure 8 shows the result for this variable based on Psych.pearson (Psych.polychoric gave slightly better results, but was hampered by warnings). This time the distinction was clearest for the percentage of variance explained. Scales with a change from unifactorial to multifactorial smaller than .5 are most likely unifactorial scales. In contrast, values above 1 increasingly indicate a multifactorial scale. Between these, the scale is most likely to be hierarchical.

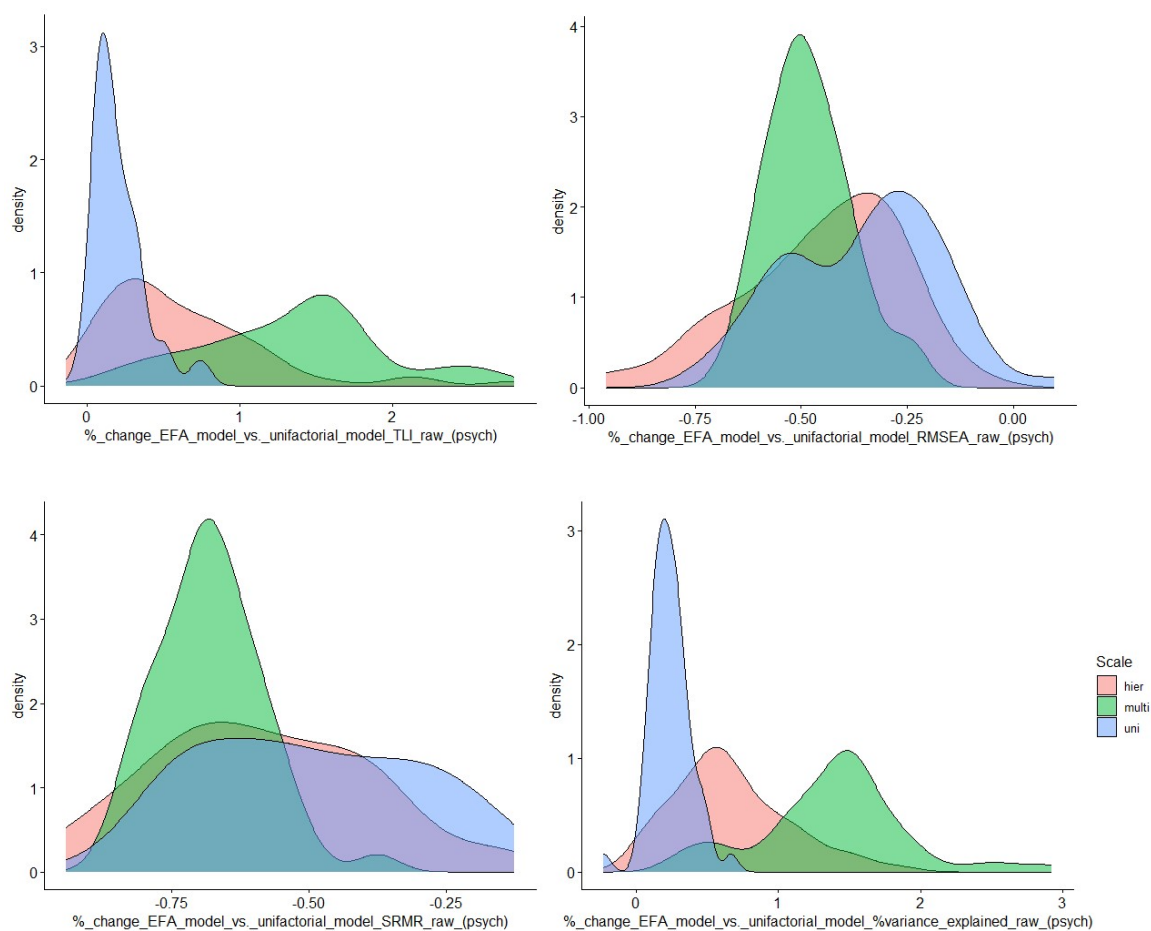


Figure 8: Values Psych.pearson for changes in TLI, RMSEA, SRMR and Percentage explained variance when the scale goes from unconstrained unifactorial to unconstrained (exploratory) multifactorial as a function of the underlying scale type.

Summary of the indicators

All in all, our analyses indicate that the Hu and Bentler standards are unrealistic for many scales in use. At the same time, there are a number of indicators that can help us determine whether a scale is unifactorial, multifactorial, or hierarchical. The indicators are not intended to replace statistical analysis of the most likely structure of a scale (based on scree plot, AIC model fit, and so on), but they can be a useful supplement to see how best to interpret a scale in practice, because not all statistical differences have practical implications. It is entirely possible that a scale may fit statistically significantly better as a hierarchical scale than as a unifactorial scale, but that there is no practical benefit to calculating component scores in addition to an overall test score.

It is also important to keep in mind that the indicators are based on studies that used an existing scale and not on studies that created a new scale. In the latter case, the fits to the model tend to be higher

due to over-fitting on the dataset on which the scale was developed. The current indicators apply only to independent studies.

Table 2 summarizes the indicators for the various scales derived from Figures 1-8.

Table 2: Indicators to see if data on a test better match answers on unifactorial, hierarchical or multifactorial scale.

Unifactorial scale	Hierarchical scale	Multifactorial scale
TLI Psych.pearson unifactorial model > .7	TLI Psych-pearson multifactorial model > .8	TLI Psych-pearson multifactorial model > .75
TLI Lavaan.ordered unifactorial model > .85	TLI Lavaan.ordered multifactorial model > .8	TLI Lavaan.ordered multifactorial model > .75
TLI Psych_polychoric unifactorial model > .6	TLI Psych.polychoric unifactorial model between .4 and .6	TLI Psych.pearson unifactorial model < .4
TLI Lavaan.MLR unifactorial model > .6	TLI Lavaan.MLR unifactorial model between .4 and .6	TLI Lavaan.MLR unifactorial model < .4
	TLI Lavaan.ordered unifactorial model between .55 and .85	TLI Lavaan.ordered unifactorial model < .55
	TLI Psych.pearson unifactorial model between .5 and .7	TLI Psych.pearson unifactorial model < .5
SRMR unifactorial model < .1	SRMR Psych multifactorial model < .07	SRMR Psych multifactorial model < .07
SRMR Lavaan.MLR multifactorial independent > .17		SRMR Lavaan.MLR multifactorial independent < .17
Correlation between factors in Psych > .4	Correlation between factors in Psych between .2 and .4	Correlation between factors in Psych < .2
Correlation between factors in Lavaan.ordered > .7	Correlation between factors in Lavaan.ordered between .4 and .7	Correlation between factors in Lavaan.ordered < .4
Change variance Psych.pearson from UNI to EFA < .5	Change variance Psych.pearson from UNI to EFA between .5 and 1.2	Change variance Psych.pearson from UNI to EFA > 1.2

Applying the guidelines to new datasets

In the previous sections we saw that guidelines other than those of Hu and Bentler (1999) can help us decide whether a scale is unifactorial, multifactorial or hierarchical, as summarized in Table 2. To see how useful these criteria are, we can apply them to new scales that were not included in the analysis.

Symptom Checklist SCL-90R

The symptom checklist SCL-90R is a 90-item self-reported inventory developed by Derogatis et al. (1976). It is used to assess psychological symptoms and psychological distress and contains 90 symptoms, which according to Derogatis et al. (1976) measure 9 factors: somatization, obsessive-compulsive thoughts, interpersonal sensitivity, depression, anxiety, hostility, phobic anxiety, paranoid ideation and psychoticism.

A dataset of 3773 teenagers from Southern China has been made available by Chen et al. (2020). Items were rated on a 5-point Likert scale (0 = never, 1 = slight, 2 = moderate, 3 = heavy, 4 = serious).

A factor analysis with 1 factor in Psych.pearson gives TLI = .769, RMSEA = .049, SRMR = .04, %variance explained = .33. The analysis in Psych.polychoric gives TLI = .724, RMSEA = .066, SRMR = .05, gives %variance explained = .42.

An EFA model with 9 factors in Psych.pearson gives TLI = .899, RMSEA = .033, SRMR = .02, # %variance explained = .43. The mean absolute correlation between the factors = .37. The same analysis in Psych.polychoric gives TLI = .855, RMSEA = .048, SRMR = .02, %variance explained = .54, mean absolute correlation between the factors = .37.

The unifactorial model in lavaan.ordered gives scaled TLI = .915, scaled RMSEA = .041, scaled SRMR = .046, %variance explained = .446. The unifactorial model in lavaan.MLR gives TLI = .915, scaled RMSEA = .041, scaled SRMR = .046, %variance explained = .446.

The multifactorial model with 9 factors is much more demanding in lavaan than in psych, because there is not only the assumption that the items group into 9 factors but also that the grouping happens as intended by the authors (i.e., measurement invariance). The analysis with Lavaan.MLR gives: scaled TLI = .836, scaled RMSEA = .032, scaled SRMR = .038, # %variance explained = .397, average absolute correlation between factors = .861. The analysis with Lavaan.ordered gives the warning: "covariance matrix of latent variables is not positive definite; use lavInspect(fit, "cov.lv") to investigate". If we nevertheless ask for the data, we get scaled TLI = .937, scaled RMSEA = .037, scaled SRMR = .041, # %variance explained = .394, average absolute correlation between factors = .867.

The multifactorial model with independent factors gives scaled TLI = .596, scaled RMSEA = .050, scaled SRMR = .308, # %variance explained = .388.

Table 3 shows how the findings fit into the indicators we derived. For the change in variance Psych.pearson, we have to calculate $(\text{variance EFA} - \text{variance UNI}) / \text{variance UNI} = (.43 - .33) / .33 = .3$.

Table 3: How well does the SCL-90R match the indicators for the different types of scales? A green ✓ indicates that the indicator applies, a red ✗ indicates that the indicator does not apply. Indicators in red are from models that generated a warning. Data from Chen et al. (2020)

Unifactorial scale		Hierarchical scale		Multifactorial scale	
TLI Psych.pearson unifactorial model > .7	✓	TLI Psych-pearson multifactorial model > .8	✓	TLI Psych-pearson multifactorial model > .75	✓
TLI Lavaan.ordered unifactorial model > .85	✓	TLI Lavaan.ordered multifactorial model > .8	✓	TLI Lavaan.ordered multifactorial model > .75	✓
TLI Psych_polychoric unifactorial model > .6	✓	TLI Psych.polychoric unifactorial model between .4 and .6	✗	TLI Psych.pearson unifactorial model < .4	✗
TLI Lavaan.MLR unifactorial model > .6	✓	TLI Lavaan.MLR unifactorial model between .4 and .6	✗	TLI Lavaan.MLR unifactorial model < .4	✗
		TLI Lavaan.ordered unifactorial model between .55 and .85	✗	TLI Lavaan.ordered unifactorial model < .55	✗
		TLI Psych.pearson unifactorial model between .5 and .7	✗	TLI Psych.pearson unifactorial model < .5	✗
SRMR unifactorial models < .1	✓	SRMR Psych multifactorial model < .07	✓	SRMR Psych multifactorial model < .07	✓
SRMR Lavaan.MLR multifactorial independent > .17	✓			SRMR Lavaan.MLR multifactorial independent < .17	✗
Correlation between factors in Psych > .4	✗	Correlation between factors in Psych between .2 and .4	✓	Correlation between factors in Psych < .2	✗
Correlation between factors in Lavaan.ordered > .7	✓	Correlation between factors in Lavaan.ordered between .4 and .7	✗	Correlation between factors in Lavaan.ordered < .4	✗
Change variance Psych.pearson from UNI to EFA < .5	✓	Change variance Psych.pearson from UNI to EFA between .5 and 1.2	✗	Change variance Psych.pearson from UNI to EFA > 1.2	✗

The SCL-90-R scale was never thought to be multifactorial, and this is reflected in the results. Other than the fact that a multifactorial model also fits the data, there is no other indicator. A more interesting question is whether there is enough evidence for a hierarchical scale rather than a unifactorial scale? Remember that the scale is supposed to measure 9 components. Looking at Table 3, we see several indicators that do not fit a hierarchical scale. In contrast, all but one of the indicators point to a unifactorial scale. The only indicator that was missed (correlation in Psych between factors > .4) was only just missed. The mean correlation was .37 for both Psych.pearson and Psych.polychoric.

Thus, for all practical purposes, the SCL-90R resembles unifactorial scales more than hierarchical scales. Little additional information can be expected from the nine factor scores. We also see this in a scree plot

(Figure 9). Although several factors have eigenvalues higher than predicted by noise and a parallel analysis suggests that up to 19 factors and 7 components can be distinguished, there is a clear elbow between factor 1 and factor 2. Apparently, responses on the SCL-90-R scale are largely determined by one factor. This could be different vulnerability to psychopathological symptoms (e.g., Caspi et al., 2014) or different response biases among participants. Regarding the latter, differences in symptom severity perception (van Wijk & Kolk., 1996), a preference for or against extremes, and differences in acquiescence bias (Fontaine et al., 2022) could contribute to this (see also Watts et al., 2024).

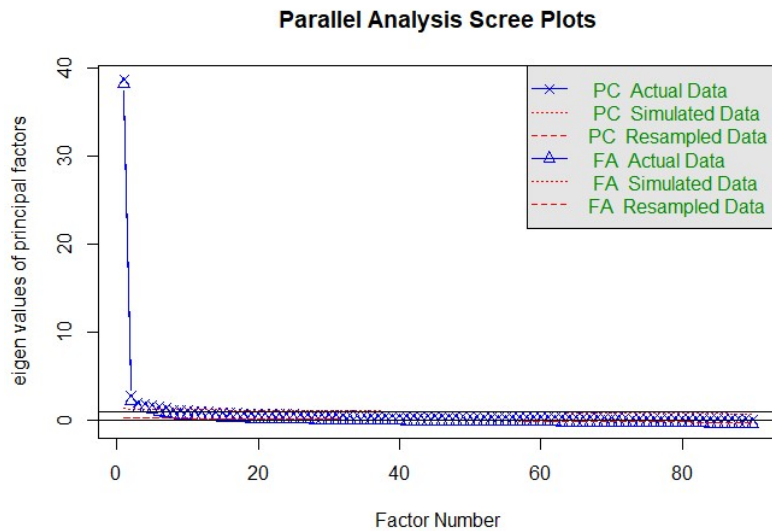


Figure 9: Scree plot of the data for Chen et al. (2020), showing the strong decrease between factor 1 and factor 2 (based on polychoric correlations).

Importantly, none of the Hu and Bentler criteria were met. Ironically, the model that comes closest (Lavaan.ordered multifactorial) came with a warning about the calculations and should be interpreted cautiously.

A very similar conclusion emerged from an analysis of another study with the SCL-90-R scale (Urbán et al., 2014) involving a community sample of 2524 participants (see the osf repository).

Domain-Specific Risk-Taking (DOSPERT) Scale

Blais and Weber (2006) published a Domain-Specific Risk-Taking (DOSPERT) scale. This estimates the degree of risk taking and/or risk perception in adults. The authors hypothesized that risk-taking would be domain-specific, such that someone willing to take financial risks need not be simultaneously inclined to take health risks. Five domains were identified, each measured by six questions: Ethical, Financial, Health/Safety, Recreational and Social.

Shou and Olney (2020) conducted a meta-analysis of DOSPERT studies and reported evidence for the main structure of the scale, with reasonable reliability of the five factor scores and factor-intercorrelations between .18 and .47.

Frey et al. (2023) provided a dataset of 3123 U.S. adults, which they also analyzed. They found reasonably good performance for the theoretical model of five intercorrelated factors, but the model "did not quite achieve a satisfactory model fit (e.g., RMSEA = .07, CFI = .80)." A bifactor model with one general factor and five orthogonal factors performed better and "approached a satisfactory fit (e.g., RMSEA = .06, CFI = .86)" (Frey et al., 2022, pp. 7-8). Based on these analyses, the authors distinguished four types (clusters) of risk takers: the cautious, the recreational adventurers, the financial gamblers and the daredevils.

An analysis of the risk taking propensities questions with our pipeline gave the following results (there were 7 response alternatives:

Unifactorial analysis

- Psych.pearson: TLI = .388, RMSEA = .12, SRMR = .11, %variance explained = .19.
- Psych.polychoric: TLI = .385, RMSEA = .147, SRMR = .13, gives %variance explained = .25.
- Lavaan.ordered: scaled TLI = .557, scaled RMSEA = .160, scaled SRMR = .144, %variance explained = .261.
- Lavaan.MLR: scaled TLI = .379, scaled RMSEA = .110, scaled SRMR = .111, %variance explained = .208.

Multifactorial analysis with correlated factors

- Psych.pearson: TLI = .847, RMSEA = .06, SRMR = .03, %variance explained = .42, average correlation between factors = .247.
- Psych.polychoric: TLI = .816, RMSEA = .08, SRMR = .04, %variance explained = .50, average correlation between factors = .249.
- Lavaan.ordered: scaled TLI = .878, scaled RMSEA = .084, scaled SRMR = .082, %variance explained = .467, average absolute correlation between factors = .352.
- Lavaan.MLR: scaled TLI = .775, scaled RMSEA = .066, scaled SRMR = .073, %variance explained = .327, average absolute correlation between factors = .317.

Multifactorial analysis with orthogonal factors

- Lavaan.MLR: scaled TLI = .698, scaled RMSEA = .077, scaled SRMR = .149, %variance explained = .334.

Table 4 shows how the findings correspond to the different indicators. The indicators clearly show that the scale is not unifactorial. It most closely resembles multifactorial scales, given that the two missed indicators were only narrowly missed (TLI Lavaan.ordered unifactorial model = .557 instead of < .55; correlation between factors in Psych = .25 instead of < .2). This suggests that risk-taking is indeed more or less independent of each other in the five domains tested. The reason the scale was not quite in line with multifactorial scales was the rather high correlation between risk-taking ethical and risk-taking health/safety (see also Shou and Olney, 2020). This suggests that it might be better to group these two factors together or rephrase some questions so that they have a smaller cross load. Some health/safety questions seem to have an ethical/legal component, such as "drinking heavily at a social gathering,"

"having unprotected sex," "driving a car without a seat belt," and "riding a motorcycle without a helmet." Replacing these items with questions purely related to health/safety may reduce correlations between the two factors.

Table 4: How well does the DOSPERT match the indicators for the different types of scales? A green ✓ indicates that the indicator applies, a red ✗ indicates that the indicator does not apply. Indicators in red are from models that generated a warning. Data from Frey et al. (2023)

Unifactorial scale		Hierarchical scale		Multifactorial scale	
TLI Psych.pearson unifactorial model > .7	✗	TLI Psych-pearson multifactorial model > .8	✓	TLI Psych-pearson multifactorial model > .75	✓
TLI Lavaan.ordered unifactorial model > .85	✗	TLI Lavaan.ordered multifactorial model > .8	✓	TLI Lavaan.ordered multifactorial model > .75	✓
TLI Psych_polychoric unifactorial model > .6	✗	TLI Psych.polychoric unifactorial model between .4 and .6	✗	TLI Psych.pearson unifactorial model < .4	✓
TLI Lavaan.MLR unifactorial model > .6	✗	TLI Lavaan.MLR unifactorial model between .4 and .6	✗	TLI Lavaan.MLR unifactorial model < .4	✓
		TLI Lavaan.ordered unifactorial model between .55 and .85	✓	TLI Lavaan.ordered unifactorial model < .55	✗
		TLI Psych.pearson unifactorial model between .5 and .7	✗	TLI Psych.pearson unifactorial model < .5	✓
SRMR unifactorial model < .1	✗	SRMR Psych multifactorial model < .07	✓	SRMR Psych multifactorial model < .07	✓
SRMR Lavaan.MLR multifactorial independent > .17	✗			SRMR Lavaan.MLR multifactorial independent < .17	✓
Correlation between factors in Psych > .4	✗	Correlation between factors in Psych between .2 and .4	✓	Correlation between factors in Psych < .2	✗
Correlation between factors in Lavaan.ordered > .7	✗	Correlation between factors in Lavaan.ordered between .4 and .7	✗	Correlation between factors in Lavaan.ordered < .4	✓
Change variance Psych.pearson from UNI to EFA < .5	✗	Change variance Psych.pearson from UNI to EFA between .5 and 1.2	✗	Change variance Psych.pearson from UNI to EFA > 1.2	✓

Difficulties in Emotion Regulation Scale (DERS)

Gratz and Roemer (2004) developed the Difficulties in Emotion Regulation Scale (DERS). They called it a multidimensional assessment because it contained six subscales measured with 36 items:

1. Non-acceptance of emotional responses
2. Difficulty engaging in goal-directed behavior
3. Impulse control difficulties
4. Lack of emotional awareness
5. Limited access to emotion regulation strategies
6. Lack of emotional clarity

Lee et al. (2016) found evidence for five factors, with lack of emotional awareness not emerging as a clear dimension (see also Sörman et al., 2022).

Valencia et al. (2024) made a dataset available of 345 participants who responded on a 5-point Likert scale. The results of the pipeline are as follows:

Unifactorial analysis

- Psych.pearson: TLI = .604, RMSEA = .124, SRMR = .13, %variance explained = .38.
- Psych.polychoric: TLI = .548, RMSEA = .155, SRMR = .14, gives %variance explained = .44.
- Lavaan.ordered: scaled TLI = .746, scaled RMSEA = .170, scaled SRMR = .144, %variance explained = .568.
- Lavaan.MLR: scaled TLI = .608, scaled RMSEA = .109, scaled SRMR = .125, %variance explained = .448.

Multifactorial analysis with correlated factors

- Psych.pearson: TLI = .915, RMSEA = .057, SRMR = .02, %variance explained = .62, average correlation between factors = .331.
- Psych.polychoric: TLI = .822, RMSEA = .097, SRMR = .03, %variance explained = .69, average correlation between factors = .343.
- Lavaan.ordered: scaled TLI = .872, scaled RMSEA = .120, scaled SRMR = .103, %variance explained = .708, average absolute correlation between factors = .613.
- Lavaan.MLR: scaled TLI = .825, scaled RMSEA = .073, scaled SRMR = .097, %variance explained = .636, average absolute correlation between factors = .553.

Multifactorial analysis with orthogonal factors

- Lavaan.MLR: scaled TLI = .676, scaled RMSEA = .099, scaled SRMR = .330, %variance explained = .613.

The indicators, summarized in Table 5, strongly point towards a hierarchical scale, the more because the only missed indicator was missed very narrowly (TLI Lavaan.MLR unifactorial model = .608, rather than between .4 and .6). This suggests that it is meaningful to look at the DERS subscores in addition to the overall score, even though the subscores are rather highly correlated.

Table 5: How well does the DERS match the indicators for the different types of scales? A green ✓ indicates that the indicator applies, a red ✗ indicates that the indicator does not apply. Indicators in red are from models that generated a warning. Data from Valencia et al. (2024)

Unifactorial scale		Hierarchical scale		Multifactorial scale	
TLI Psych.pearson unifactorial model > .7	✗	TLI Psych-pearson multifactorial model > .8	✓	TLI Psych-pearson multifactorial model > .75	✓
TLI Lavaan.ordered unifactorial model > .85	✗	TLI Lavaan.ordered multifactorial model > .8	✓	TLI Lavaan.ordered multifactorial model > .75	✓
TLI Psych_polychoric unifactorial model > .6	✗	TLI Psych.polychoric unifactorial model between .4 and .6	✓	TLI Psych.pearson unifactorial model < .4	✗
TLI Lavaan.MLR unifactorial model > .6	✓	TLI Lavaan.MLR unifactorial model between .4 and .6	✗	TLI Lavaan.MLR unifactorial model < .4	✗
		TLI Lavaan.ordered unifactorial model between .55 and .85	✓	TLI Lavaan.ordered unifactorial model < .55	✗
		TLI Psych.pearson unifactorial model between .5 and .7	✓	TLI Psych.pearson unifactorial model < .5	✗
SRMR unifactorial model < .1	✗	SRMR Psych multifactorial model < .07	✓	SRMR Psych multifactorial model < .07	✓
SRMR Lavaan.MLR multifactorial independent > .17	✓			SRMR Lavaan.MLR multifactorial independent < .17	✗
Correlation between factors in Psych > .4	✗	Correlation between factors in Psych between .2 and .4	✓	Correlation between factors in Psych < .2	✗
Correlation between factors in Lavaan.ordered > .7	✗	Correlation between factors in Lavaan.ordered between .4 and .7	✓	Correlation between factors in Lavaan.ordered < .4	✗
Change variance Psych.pearson from UNI to EFA < .5	✗	Change variance Psych.pearson from UNI to EFA between .5 and 1.2	✓	Change variance Psych.pearson from UNI to EFA > 1.2	✗

Discussion

Test developers have self-imposed the findings from a simulation study by Hu and Bentler (1999) as a general quality stamp.³ Tests with goodness of fit measures of CFI > .95, TLI > .95, RMSEA < .06 and SRMR < .08 are considered to be the best of the best.

Two assumptions underlie the practice: (1) that Hu and Bentler's criteria apply to all tests and (2) that extremes represent the best situation. The latter need not be true, as evidenced by the temperature of the air around us. Humans function optimally at temperatures between 18 and 30 degrees Celsius. Colder temperatures are less good and require adaptations. But so do hotter temperatures. There is a sweet spot (or region), beyond which performance declines and where too much of a desired quality becomes bad again. Curvilinear functions are everywhere in biology (there is also an optimal level of food intake) and in psychology (there are optimal levels of performance motivation, perfectionism, and so on ; Grant & Schwartz, 2011; Pierce & Aguinis, 2013; Hornsey et al., 2018; Stavrova & Ren, 2021).

There is evidence that very high values of CFI and TLI and very low values of RMSEA and SRMR are not necessarily good. It favors the top-down construction of scales that measure narrow constructs with a small number of items (Cooper, 2024). A test that works bottom-up by first writing questions that map the entire problem space and then looking at the correlations present between items to understand the structure is unlikely to result in a scale with the desired quality approval, as discussed in the Introduction.

The exclusive focus on the Hu and Bentler (1999) norms has arguably also led to a proliferation of bifactor models because they often provide higher fit measures, even if such models do not lead to test scores that are more useful in practice (Greene et al., 2019, 2022; but see Bader & Moshagen).

Finally, Hu and Bentler's criteria have sparked a rush to develop more powerful analysis techniques. As this article shows, there are large differences between the estimates produced by existing techniques and software packages. The undisputed leader at present is DWLS and related techniques (Muthén & Muthén, 2007). This technique makes it possible to approximate the Hu and Bentler standards (Figures 1-3). Unfortunately, the technique often results in warnings with real data, indicating that the calculations were not quite correct (to the despair of the researchers involved) and occasionally resulting in poor estimates (as befits a good warning signal). It is also wrong to use the Hu and Bentler criteria for a technique that is more powerful than the one from which the criteria were derived. Xia and Yang (2019) showed that if you want to stay within Hu and Bentler's philosophy, you need stricter criteria for DWLS than Hu and Bentler's (making it equally impossible for real tests to pass them).

An alternative approach is not to rely on simulations, but to compile practice-based distributions. What are the values obtained with existing tests? What values allow us to choose between unifactorial, hierarchical and multifactorial scales? Surprisingly, I could not find such an article, possibly because datasets are only now becoming available in sufficient numbers due to the open science movement (Weston et al., 2019).

An analysis of 161 datasets shows that there are indeed regions of high probability for TLI, RMSEA and SRMR (Figures 1-3) and that these regions are largely outside the Hu and Bentler regions (also depending on the method used). Further analysis shows that different measures have different distributions for

³ McNeish (2023) notices that Hu & Bentler (1999) is one of the most cited papers in psychology.

unifactorial, hierarchical and multifactorial scales, making it possible to use them as indicators of scale type (Figures 5-8). Interestingly, RMSEA is not among these indicators, as it seems to make little difference between the three types of scales.

Based on the distributions, indicators that differentiate between scale types were selected (Table 2). These indicators were used to find the best fitting scale type for three tests that were not included in the initial analysis phase, with satisfactory results. When using these indicators, it is good to remember three things. First, they are based on independent (validation) studies. They may not work if one tries to use them for stimulus selection in test development. Second, they are indicators because there is often considerable overlap between distributions, making it unwise to consider the values as strict boundaries. It is better to keep the complete distributions from Figures 4-8 in mind when interpreting them. Finally, the indicators are not intended to replace statistical model comparison, but to enhance them by providing a practice-oriented view in addition to a purely statistical one.

Limitations

The indicators presented here have two major limitations. First, they are based on a fairly small set of studies. They show the big picture, but a few thousand studies with a greater variety of scales would provide more detail and better values for the indicators.

Second, I have tried only a few indicators. It is very likely that there are others and better ones. One of the alternatives I explored was the use of a bifactorial analysis, based on the assumption that the bifactorial framework can be seen as an overarching structure, with unifactorial scales loading high on the general factor affecting all items, and multifactorial scales loading high on the specific items (Dueber & Toland, 2023). Hierarchical scales then sit somewhere in between. Indeed, analyzing explained common variance (ECV) indices confirmed the promise of this approach, with higher overall ECV values for unifactorial than for multifactorial designs (and vice versa for ECV values for specific factors), but the distributions were not as clear as those in the current article. The allocation of variance to the general and specific factors appears to be less stable than hoped. However, further refinements may be possible.

As such, this article is only a first attempt to develop new indicators for scale evaluation.

Data availability

The datasets and the R code used for the present article are available at https://osf.io/e3dmh/?view_only=2196558e8c3e4ae3bfff6b60bf0b9464.

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